

Paired Kidney Exchange

Kidney Transplant Compatibility System

How the system identifies viable kidney swaps between incompatible donor–recipient pairs, how it chooses among them, how a proposed swap becomes a committed one, and what the system does not yet do.

Scope Paired exchange subsystem — pool, matching, allocation policy, commitment workflow

Audience Transplant clinicians · programme coordinators · academic supervision · hospital IT

Status Implemented and tested. Commitment workflow live; not yet used on real patients

Reference data HLA frequencies `grifoni-colombo-2018-v1` (n=714, Colombo)

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1 Summary

A patient with a willing living donor who cannot donate to them is not out of options. If two such pairs are incompatible in complementary ways, each donor may be able to give to the other's recipient. The paired exchange feature finds those opportunities across every participating hospital, ranks them, and manages the agreement between the clinicians involved.

The feature does five things:

STAGE	WHAT HAPPENS
Assemble the pool	Every registered donor-recipient pair that is verified, active, and <i>internally incompatible</i> . Pairs who can already transplant directly are excluded — they do not need a swap.
Test every direction	For each ordered pair of pairs, decide whether that donor could give to that recipient: blood group, HLA mismatch, and a virtual crossmatch against the recipient's antibodies.
Find the closed loops	Enumerate every two-way and three-way cycle in which each donor gives to the next recipient and the last gives back to the first.
Choose the best set	Solve a binary integer programme that picks the highest-scoring set of non-overlapping cycles under one of four allocation policies.
Commit it	A coordinator proposes a cycle; each pair's own doctor accepts; only when all have accepted are the donors reserved and the cycle locked.

What it is, and what it is not

It is a **decision-support tool**. Every clinical judgement it makes is made from data a human verified first, and the swap it proposes cannot proceed until every clinician involved has agreed to their own part of it. It performs a virtual crossmatch, not a real one — an actual laboratory crossmatch remains mandatory before any transplant.

It is **not** an allocation authority, and it does not act on its own. It reserves a donor only after unanimous human acceptance, and it can be unwound at any point before transplant.

VERIFICATION STATUS

The exchange subsystem carries **58 backend and 34 frontend automated tests**, part of 831 across the whole system, all passing. The cycle enumerator is checked against an independent brute-force implementation over 500 randomly generated graphs. Allocation is deterministic: a tie between two equally-scoring swaps resolves the same way every time, and that is verified over 100 consecutive runs.

2 The clinical problem

Roughly a third of willing living kidney donors are turned away because they are incompatible with the person they came to donate for — most often on blood group, sometimes because the recipient carries antibodies against the donor's tissue

type. The organ is available, the donor is willing, and the transplant cannot happen.

Paired exchange resolves this by looking at incompatibility as a property of the *combination* rather than of the donor. If pair A's donor is blood group B and pair B's donor is blood group A, and each recipient needs what the other's donor has, the two pairs can swap.

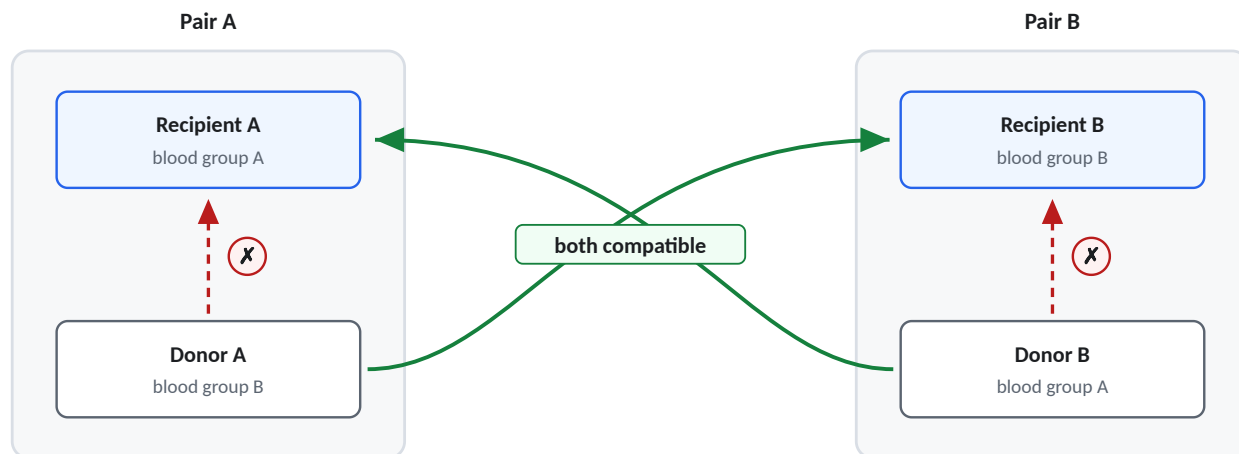


Figure 1 — A two-way exchange. Neither donor can give within their own pair (dashed, red). Crossed over, both directions clear every compatibility test. Two transplants that could not otherwise happen.

The same logic extends to three pairs in a closed loop: A's donor gives to B's recipient, B's donor to C's recipient, and C's donor back to A's recipient. Three-way cycles find matches that no two-way swap can reach — but every nephrectomy in a cycle must happen simultaneously, so that a donor cannot withdraw after their own recipient has already received a kidney. That surgical constraint, not a software one, is why the system stops at three.

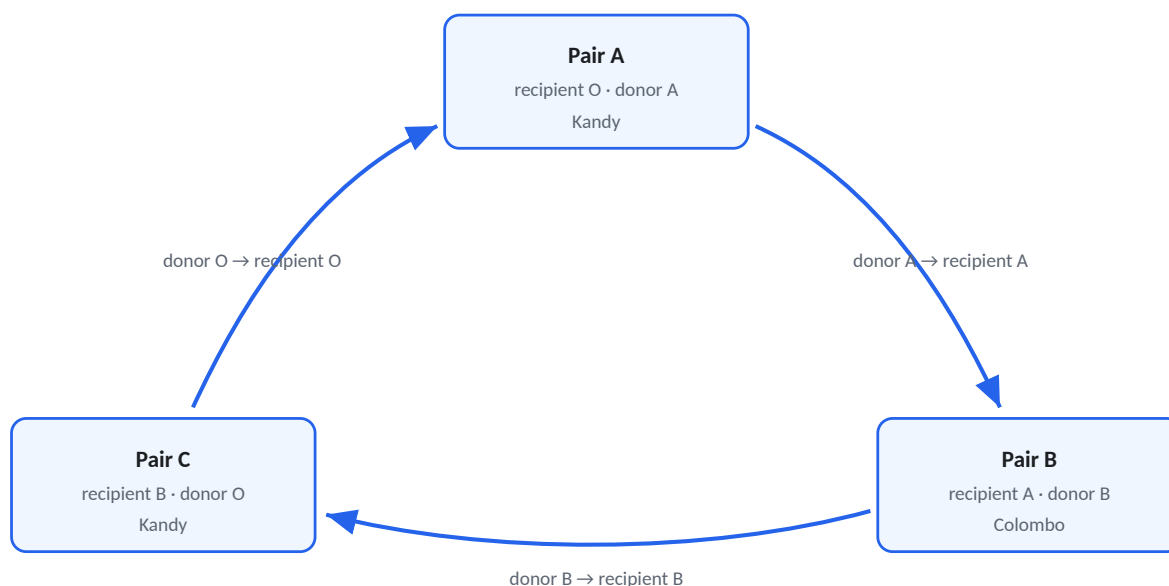


Figure 2 — A three-way cycle. Each donor gives to the next pair's recipient and the loop closes. Cycles may span hospitals, which is why the matcher searches the national pool rather than one centre's own patients.

3 Which pairs enter the pool

A pair is eligible only when a human has verified every piece of data the matcher will rely on. Five separate verification flags must all be set:

REQUIREMENT	WHY
Donor HLA typing verified	Mismatch counting is meaningless on unconfirmed typing
Recipient HLA typing verified	
Recipient antibody profile verified	The virtual crossmatch is only as good as the bead-chart transcription behind it
Donor details verified	Blood group and demographics drive both the ABO gate and the donor-quality score
Recipient details verified	

On top of those, the donor must be **available** (not reserved, under work-up, withdrawn, medically unfit, transplanted or deceased), neither record may be soft-deleted, and the donor must have a nominated intended recipient — an altruistic donor with no recipient is not a pair and cannot participate in a cycle.

Two deliberate exclusions

Two further groups are filtered out after the compatibility of the pair with itself is computed:

- **Directly compatible pairs are removed.** If the donor can already give to their own recipient, they should simply proceed to transplant. Including them would let the optimiser break up a working pair to improve someone else's match.
- **Pairs with incomplete HLA typing are removed.** The mismatch calculator assigns the worst possible score to a missing locus. That biases toward rejection but does not guarantee it — so an incompletely typed pair could previously slip into the pool on guessed data with nothing flagging it. They are now excluded outright.

KNOWN GAP

A pair that is part of a *proposed but not yet fully accepted* cycle remains in the pool and will keep being suggested in new matches. The safeguard is at the point of commitment — a second proposal for the same donor is refused — but the coordinator is not warned until they try. Pairs leave the pool only once a cycle is fully accepted and their donor becomes reserved.

4 Deciding that a donor can give

For every ordered pair of pool members the system asks a single question: *could this donor give to that recipient?* Three gates must all pass. The test is directional — that A's donor suits B's recipient says nothing about the reverse.

4.1 Blood group

Standard ABO compatibility, following OPTN policy. Rhesus factor is displayed for completeness but plays no part in the decision: RhD is not expressed on renal vascular endothelium in a way that drives rejection, unlike its role in transfusion.

RECIPIENT	ACCEPTABLE DONOR GROUPS	NOTE
O	O	Hardest to match — universal donor, restricted recipient
A	A, O	
B	B, O	
AB	A, B, AB, O	Universal recipient

4.2 HLA mismatch

Mismatches are counted across three loci — **A, B and DRB1** — as the number of donor alleles the recipient does not also carry. With two alleles per locus the count runs from 0 to 6. A pair is rejected at **6 mismatches**, inclusive: since 6 is also the maximum the count can reach, a strict "greater than 6" rule would never fire at all.

A locus with missing typing is scored at the worst case of 2 and clears a data-completeness flag — which, per §3, now removes the pair from the pool entirely rather than letting it compete on assumed values.

4.3 Virtual crossmatch

The recipient's bead-specificity profile is checked against the donor's HLA antigens. Any antibody above a mean fluorescence intensity of 1,000 that targets an antigen the donor carries is a donor-specific antibody, graded into three bands:

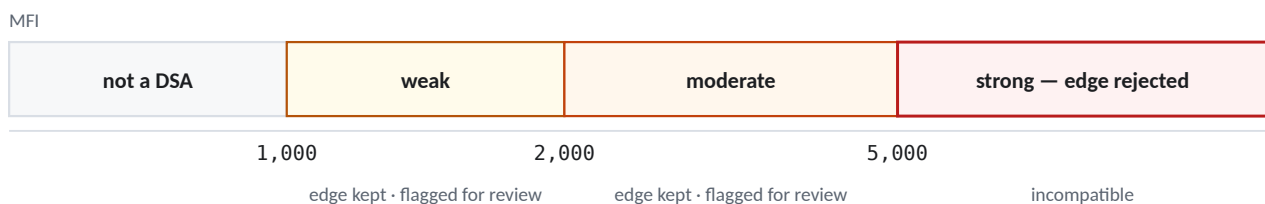


Figure 3 — Donor-specific antibody bands. Only a strong DSA blocks a match. Weak and moderate antibodies leave the swap on the table but mark it as requiring desensitisation review — the interface draws those connections as dashed lines.

4.4 What is deliberately not tested

- **cPRA / sensitisation** — a population-level measure of how hard a patient is to match, not a property of a specific donor–recipient pairing. It informs allocation priority (§6), not eligibility.
- **Physical crossmatch** — requires same-day serum from both parties. It cannot exist for a hypothetical swap. **A laboratory crossmatch remains mandatory before any transplant proceeds.**
- **Age, size and comorbidity matching** — not modelled. Donor quality enters only through the LKDPI score used by one optional allocation policy.

5 Choosing the set of swaps

Testing every direction yields a directed graph: each pair is a node, each viable donation an arrow. A swap is a closed loop in that graph. The remaining problem is choosing *which* loops to run, given that no pair can be in two of them.

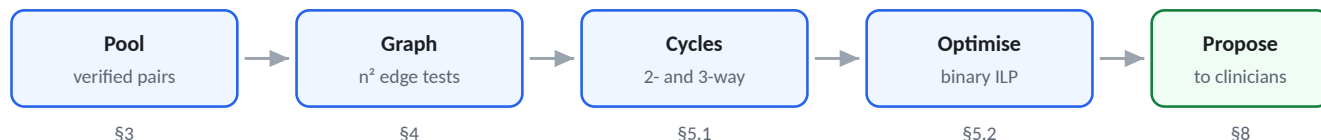


Figure 4 — The matching pipeline. Every stage is recomputed on demand; nothing is cached between requests, so the answer always reflects the pool as it stands right now.

5.1 Enumerating the cycles

Two-way cycles are found by testing each unordered pair of nodes for arrows in both directions. Three-way cycles are found by walking outward from each existing edge: for every edge $i \rightarrow j$ and every onward neighbour k of j , the cycle closes if an edge $k \rightarrow i$ exists. Each cycle is then rotated to begin at its lowest-numbered pair so the same loop is never counted twice.

Rotations are treated as duplicates; **reversals are not**. A three-way cycle running $A \rightarrow B \rightarrow C$ and the one running $A \rightarrow C \rightarrow B$ use entirely different donations and are genuinely different clinical proposals.

5.2 The optimisation

Every candidate cycle becomes a yes/no decision variable. The solver maximises the total score of the cycles it turns on, subject to two constraints:

```
maximise     $\sum wc \cdot xc$                                  $xc \in \{0, 1\}$ , one per candidate cycle
subject to   $\sum_{c \ni \text{pair}} xc \leq 1$     for every pair in the pool
             $\sum_{c \ni \text{patient}} xc \leq 1$     for every patient
```

The first constraint stops one pair being committed to two swaps. The second exists because a patient registered with *two* willing donors appears as two separate pool entries — without it, the same person could be scheduled to receive two kidneys.

The problem is solved with CBC, the open-source integer programming solver, through the PuLP modelling library. If the solver returns anything other than a proven optimum the request fails loudly rather than presenting a partial answer as a complete one.

WHY AN EXACT SOLVER

A greedy heuristic — take the best cycle, remove its pairs, repeat — is simpler but can leave transplants on the table. Choosing a two-way swap first may block a three-way that would have served more patients. At the pool sizes this programme will see, an exact optimum is affordable: graph construction and solving together take about 5.6 seconds at 300 pairs.

6 The four allocation policies

"Best set of swaps" is a policy question, not a technical one. The system implements four scoring rules and makes the choice explicit rather than burying it.

POLICY	CYCLE SCORE	OPTIMISES FOR
Maximum transplants	Number of pairs in the cycle	Sheer volume — the most people transplanted, regardless of who
Maximum quality	Σ (6 – mismatches) over the cycle's donations	Immunological closeness, which correlates with long-term graft survival
Equity weighted	Σ per patient of 1.0 + cPRA fraction + waiting fraction	Reaching hard-to-match and long-waiting patients
Donor quality (LKDPI)	$\Sigma \max(0, 100 - \text{LKDPI})$ over the donations	Predicted graft longevity from donor characteristics

Under equity weighting, waiting time is measured from the recipient's dialysis start date, normalised over three years and capped. Where that date has not been recorded the system falls back to the date the record was created and **says so explicitly** in the response, so a proxy is never mistaken for a fact.

Ties are broken deterministically

Under maximum-transplants scoring in particular, many different combinations of swaps produce identical totals. Left to the solver, which patients get transplanted among equally-scoring options would be arbitrary — unacceptable in an allocation tool, even an advisory one.

Each cycle's score is therefore scaled by one million and a waiting-time term of at most 300 is added:

```
tie_broken_weight = round(policy_score × 1,000,000) + round(total_wait_fraction × 100)
```

Because the tie-break can never exceed 300, it decides only between cycles whose real scores differ by less than 0.0003 — differences with no meaning. The rule it encodes is stateable and defensible: **among equally good solutions, prefer the one serving the longest-waiting patients**. The score shown in the interface remains the unscaled policy value.

VERIFIED

Two tied two-way swaps, one containing a patient waiting 1,000 days: the longer-waiting cycle is selected on all 100 consecutive runs.

7 A worked example

A reproducible research harness generates a synthetic pool and scores every policy against every other policy's yardstick. Run with a fixed seed, it produces a 40-pair pool with **212 viable donations** between them.

All four policies transplant the same number of people — **6 cycles, 12 pairs**. What differs is *which* twelve.

POLICY	MISMATCH QUALITY	EQUITY SCORE	LKDPI QUALITY	CPRA MATCHED	CPRA UNMATCHED
Maximum transplants	29.0	25.5	900.6	34.2%	25.5%
Maximum quality	34.0	23.4	882.0	31.0%	26.8%
Equity weighted	26.0	25.5	859.8	34.2%	25.5%
Donor quality (LKDPI)	32.0	24.3	951.3	31.0%	26.8%

Each policy wins its own column, which is the least one can ask of it. The clinically interesting result is in the last two: the policies that reach the *most sensitised* patients — average cPRA 34.2% among those matched, against 25.5% among those left behind — are maximum-transplants and equity weighting. Maximising immunological quality inverts that relationship, matching cPRA 31.0% while leaving 26.8% unmatched. **Chasing the best-matched kidneys quietly leaves the hardest-to-match patients in the pool.**

In this particular pool equity weighting selects exactly the same twelve pairs as maximum transplants; maximum quality uniquely reaches two pairs the others miss while dropping three that they serve. That divergence, not the headline count, is what the comparison view exists to show.

A hand-checkable case

The integration test suite seeds the textbook two-way swap: two pairs with identical HLA typing (A 01/02, B 07/08, DRB1 03/04), each internally ABO-incompatible, each compatible crossed over. The system returns 2 nodes, 2 edges, one selected cycle of weight 2.0. Small enough to verify by hand, and it runs on every commit.

CAVEAT ON THESE FIGURES

This is a synthetic pool, not Sri Lankan registry data. It demonstrates that the policies diverge and in which direction — it does not predict what they would do here. The comparison should be regenerated against real pool data before any policy is adopted as programme practice, and it should be regenerated now in any case: these numbers predate two recent changes to how cPRA is computed.

8 From proposal to commitment

Finding a swap is the easy half. A cycle spans hospitals, and no clinician may commit another centre's patient. The commitment workflow is built around that constraint.

8.1 The sequence

1. A coordinator reviews the selected cycles and **proposes** one. The server re-runs the match and verifies the cycle really is in the current selected set — a proposal cannot be fabricated by a crafted request.
2. The clinical picture is **snapshotted** onto the proposal: every node, every donation, the mismatch counts, DSA results and LKDPI scores as computed at that moment, together with the version of the HLA frequency table and the DSA band boundaries then in force.
3. Each pair's **own doctor** is asked to decide. A pending-decision badge appears in their navigation, refreshed every 30 seconds.
4. Any single **decline** ends the whole proposal, with an optional reason recorded.
5. Only on the **last acceptance** does the cycle lock: every donor moves from available to reserved in one transaction, and the pairs leave the pool.

WHY SNAPSHOT

A cycle proposed on Monday and accepted on Thursday must show the clinical picture it was accepted on — not a re-run against a pool that has since changed, or against reference data that has since been updated. The snapshot is what makes the record of a decision meaningful afterwards.

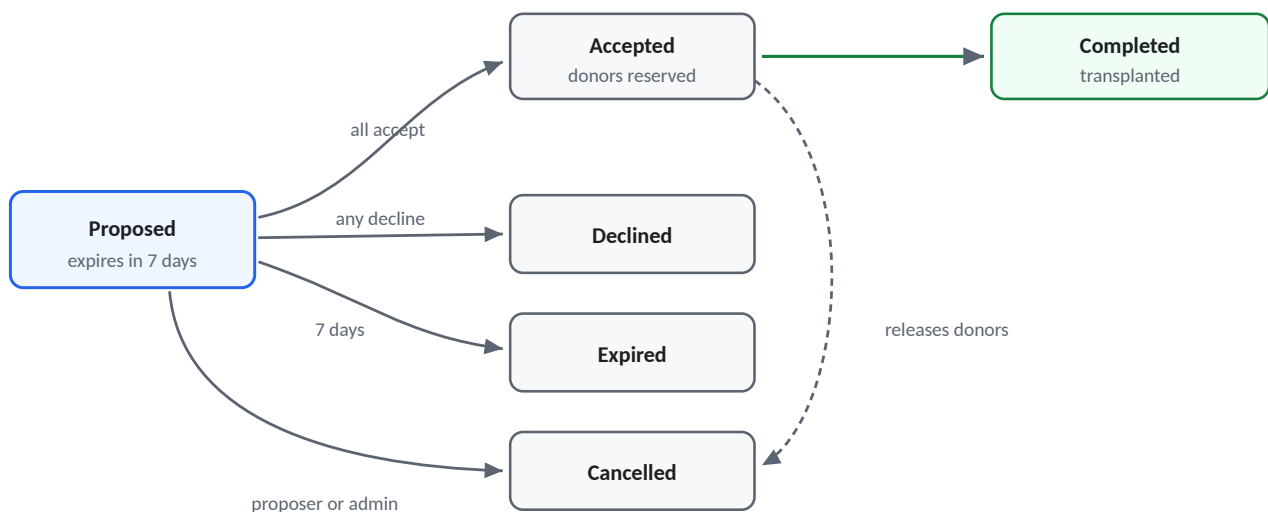


Figure 5 — Proposal states. Declined, expired and cancelled are terminal. Cancelling an already-accepted cycle releases the reserved donors back into the pool, so a commitment is never irreversible before surgery.

8.2 Preventing double-booking

The rule that no pair may sit in two live proposals is enforced at three depths, deliberately:

LAYER	ROLE
Database index	A partial uniqueness constraint on the donor across open proposals. This is the actual guarantee — it holds even if the code above it is wrong.
Advisory lock	Serialises simultaneous proposal creation so two coordinators clicking at once cannot race.
Pre-check	Produces the friendly "already committed to proposal #12" message rather than a database error.

Exactly one function in the codebase is permitted to change a proposal's status, and it updates the open/closed marker on every constituent pair in the same transaction — so the two can never disagree.

8.3 Expiry

Proposals expire after seven days so that a pair is never blocked indefinitely by a decision nobody made. Expiry is applied both lazily, whenever a proposal is read, and by a sweep when the service restarts.

OPERATIONAL NOTE

There is currently no email or SMS notification. A doctor learns of a pending decision from the badge in the application, and the proposal screen surfaces the other clinicians' email addresses for manual follow-up. For a cross-hospital programme this should be revisited once proposals routinely sit unanswered for more than a working day.

9 Why a pair was not matched

The most common question a coordinator asks is not "who matched?" but "why didn't my patient?" Every pair left out of the selected set receives an explanation, derived from information gathered while the graph was being built.

VERDICT	MEANING	WHAT IT POINTS TO
No donor out	This pair's donor is incompatible with every recipient in the pool	Pool composition. Nothing about this patient can change it
No donor in	No donor in the pool can give to this recipient	The sensitisation signal. Consider desensitisation or a wider referral
No reciprocal path	Compatible with others, but sits in no closed two- or three-way loop	More pairs would help; a longer chain would help but is not supported
Lost to overlap	Appears in candidate cycles, but the optimiser preferred an overlapping alternative	This patient was viable. Try another policy, or the comparison view

Each verdict is accompanied by counts of *why* the blocked directions failed — for example "incompatible with 11 on blood group, 2 on strong DSA, 1 on mismatch". Each rejected donation is attributed to a single reason in fixed priority, so the counts add up rather than double-counting.

The explanations are aggregate by design. At a 300-pair pool the full detail would be 90,000 individual results; the counts convey the same clinical message in a form a person can read.

The same data drives a **hard-to-match worklist**, ranking unmatched pairs by waiting time with their cPRA and verdict alongside. That list is the practical output for a desensitisation programme: it names the patients for whom no swap exists and says why.

10 Using the feature

The exchange screen offers three views of the same pool.

VIEW	SHOWS	USE WHEN
Single policy	Pool size, cycles selected, pairs transplanted; the graph; a card per selected cycle with a Propose action	Routine review under an agreed policy
Compare	Every candidate cycle with badges showing which policies chose it, sorted by agreement	Deciding whether a swap is robust or an artefact of one scoring rule
Hard to match	Unmatched pairs by waiting time, with cPRA and the reason they were excluded	Planning desensitisation or onward referral

A cycle is presented as a **card** — a vertical chain of pairs showing each recipient's and donor's blood group, the hospital, and the mismatch and DSA status of every donation. Cards are the primary representation at every screen size. The circular graph is a supplementary desktop view, available on a phone through a "View graph" control.

The whole feature is usable on a phone: the navigation collapses to a bottom bar, cycle cards stack, and the accept/decline controls sit in a fixed action bar above the safe area. Workflow states — pending, accepted, declined — are shown in neutral styling, never in the clinical colour palette, so that green never means two different things on the same screen.

The coordinator's journey

1. Open **Paired exchange**. Review the selected cycles under the programme's chosen policy.
2. Switch to **Compare** to see which cycles every policy agrees on. Those are the safe ones to act on first.
3. Choose a cycle and press **Propose this cycle**.
4. Each donor's doctor sees a badge on their Exchange tab and opens the inbox, which separates "Awaiting my decision" from "All open proposals".
5. They review the snapshot — the full chain, every mismatch count, every DSA flag — and accept or decline with a reason.
6. When the last acceptance lands, the proposal reads **Accepted** — **locked** and the donors are reserved.
7. Work-up and a real laboratory crossmatch proceed outside the system.

11 Reference data and provenance

Every clinical constant is traceable to a citation held in the code beside the number it justifies.

11.1 HLA antigen frequencies — Sri Lankan

cPRA depends on how common each HLA antigen is in the *local* donor population. Using a European or North American table would misstate how hard a Sri Lankan patient is to match.

Source	Grifoni et al., <i>Human Immunology</i> 2018;79(2):87–88 · PMID 29289740 · AFND population 3423
Cohort	714 healthy adult blood-bank donors, Colombo, drawn from all 26 districts
Coverage	97 antigens across A, B, C, DPB1, DQB1 and DRB1
Method	Computed by direct counting from the raw ambiguity-resolved genotype export, not transcribed from the paper's tables
Version	grifoni-colombo-2018-v1

Molecular alleles are collapsed to serological designations using the same convention as the rest of the application, so A*33:03 becomes A33. The values are carrier (phenotype) frequencies, which is the correct form for cPRA's union calculation. This table replaced an earlier approach that derived frequencies from the application's own database — a biased sample with a denominator error on top.

STATED LIMITATION — LINKAGE DISEQUILIBRIUM

cPRA is computed assuming antigens occur independently. They do not: HLA is inherited in haplotypes, so a patient sensitised to one antigen is more likely to be sensitised to those commonly inherited alongside it. This **inflates** computed cPRA. The correction requires the study's haplotype table, which could not be obtained; the code marks the exact point where a haplotype-aware step would attach. Note also that several DPB1 entries rest on a single observation in 714.

11.2 Donor quality — LKDPI

The Living Kidney Donor Profile Index (Massie et al., *American Journal of Transplantation* 2016;16(7):2077–2084) predicts graft survival from thirteen donor and donor–recipient characteristics. All thirteen coefficients were checked against the published source.

USE WITH CAUTION

LKDPI's discrimination is modest and its external validation is weak: a C-statistic of 0.59 in the derivation cohort of 36,025, falling to **0.55** in both European (n=416) and Canadian (n=645) validations, with a Japanese cohort (n=133) finding no association at all. **It has never been validated in any South Asian population.** The band boundaries used in the interface are a project convention for presentation, not clinical policy. This is why LKDPI informs one optional policy rather than gating any decision.

11.3 Other tables

TABLE	SOURCE AND VALUE
ABO compatibility	OPTN blood group compatibility policy
DSA thresholds	Floor 1,000 MFI; bands at 1,000 / 2,000 / 5,000; only strong halts
cPRA sensitisation cutoff	2,000 MFI — deliberately <i>not</i> unified with the 1,000 DSA floor; they answer different questions
Mismatch loci	A, B, DRB1; rejection at 6 of 6

12 Safeguards, audit and privacy

12.1 What the system will not let you do

- **Match on unverified data.** Five verification flags gate pool entry; a pair whose antibody profile has not been confirmed by a clinician is simply not there.
- **Commit another hospital's patient.** Only the doctor who owns a pair may accept it. Not even an administrator can accept on their behalf — an administrator may only cancel.
- **Double-book a donor.** Enforced by a database constraint, not by application logic alone.
- **Reserve a donor on a maybe.** Reservation happens on unanimous acceptance, never at proposal.
- **Move a donor through an illegal state change.** A transplanted or deceased donor cannot return to available — a rule that exists specifically to stop a ruled-out donor re-entering the exchange pool.
- **Silently return a partial answer.** A solver that fails to prove optimality raises an error rather than displaying a shorter list of swaps as though it were complete.

12.2 Audit

Every consequential action writes an entry to a **hash-chained** audit log: each record includes a hash of its predecessor, so removing or altering a historic entry breaks the chain and is detectable. Records are ordered by a database sequence rather than a timestamp, so a clock adjustment cannot reorder them, and a gap in the sequence reveals a deleted row. Concurrent writers are serialised by an advisory lock so the chain cannot fork.

Actions recorded for exchange: running a match, comparing policies, viewing the hard-to-match worklist, proposing a cycle, each individual acceptance and decline, locking, cancelling and expiring. The log is tamper-evident rather than tamper-proof — an honest description worth repeating to any reviewer who asks.

12.3 Privacy

The exchange endpoints are deliberately not restricted to the requesting doctor's own patients — a paired exchange only works if it can see across every participating hospital's pool. To make that safe, the exchange responses carry **no identifying information beyond blood group**: no names, no dates of birth, no national identity numbers. Those remain behind the owner-scoped patient and donor endpoints. What a coordinator sees of another hospital's pair is a blood group, a hospital name, and the contact details of the responsible clinician.

13 Known limitations

Stated plainly, because a decision-support tool whose limits are unclear is more dangerous than one that does less.

13.1 Clinical scope

LIMITATION	CONSEQUENCE
Cycles capped at three	Longer loops that would serve more patients are never found. The constraint is surgical — simultaneous nephrectomies — not computational
No altruistic donors or chains	A non-directed donor cannot start a chain. Only closed loops between paired donors are modelled
Virtual crossmatch only	A physical crossmatch is still mandatory. The system can only rule pairs out, never in
cPRA assumes independence	Overstates how sensitised a patient is; see §11.1
LKDPI unvalidated locally	Near-chance discrimination externally, never tested in South Asia; see §11.2
Rh factor unused	Correct for kidney transplantation, but it is displayed, which may mislead
No paediatric or size priority	Common in national allocation schemes; not modelled here

13.2 Operational

- **No notifications.** Pending decisions surface only as an in-application badge. A doctor who does not open the system does not learn a decision is waiting.
- **Pairs in a pending proposal stay in the pool** and will be re-suggested; the conflict only appears when a second proposal is attempted.
- **Waiting time falls back to record-creation date** where a dialysis start date has not been entered. Disclosed per pair, but it means equity weighting is only as good as the data entered.
- **No completion step.** The workflow ends at "accepted and locked"; recording the actual transplant is still outside the system.
- **Performance is characterised only to about 300 pairs** (~5.6 seconds). Beyond that, cycle enumeration growth has not been measured.

13.3 Known defects at the time of writing

DEFECT	STATUS
Under equity weighting, the cPRA contribution may not resolve antigen names in the combined format the bead charts produce, so some sensitised patients' priority may be understated	Flagged, needs confirmation — no test currently covers it
A recent correction to cPRA de-duplication shipped without a version bump, so historical recorded cPRA values are not comparable with new ones	Disclosure needed
Accept, decline and cancel actions have no error handler; a failed request reverts the screen with no message	Fix before clinical use
No confirmation dialog before accepting or declining an irreversible commitment	Fix before clinical use
The solver library emits deprecation warnings; the next major version will break it and the dependency is not pinned upward	Pin the version
A code comment still states that no dialysis start date field exists; it was added	Documentation drift

14 Technical appendix

14.1 Endpoints

ENDPOINT	PURPOSE AND AUTHORISATION
GET /exchange/match?policy=	Nodes, edges, selected cycles, explanations. Any authenticated doctor; not scoped to their own patients
GET /exchange/match/compare	All four policies over one shared enumeration
GET /exchange/hard-to-match	Unmatched pairs ranked by waiting time
POST /exchange/proposals	Create. 400 if the cycle is not currently selected; 409 on overlap
GET /exchange/proposals?mine=	Inbox, or all open proposals
GET /exchange/proposals/{id}	Full snapshot. Visible to every participant
POST .../pairs/{id}/decision	Accept or decline. Owning doctor only — 403 otherwise
POST .../{id}/cancel	Proposer or administrator only

14.2 Data model

Two tables support the workflow, separating the immutable from the mutable. **Exchange proposals** hold the policy, the score, the JSON clinical snapshot and the expiry. **Exchange proposal pairs** hold one row per pair with its owning doctor, its decision, who decided and when, any decline reason, and the open/closed marker the uniqueness constraint depends on.

Notably there is no "exchange" table for match results — a match is computed on demand and never stored. Only a proposal, being a human commitment, is persisted.

14.3 Performance

Graph build and solve, 300-pair pool	~5.6 s, moved off the request thread
Edge scoring	$O(n^2)$ — every ordered pair tested
Three-way cycle enumeration	$O(E \cdot d)$, edge-driven rather than scanning all triples
All four policies together	$\approx 1.3\times$ a single policy — enumeration is shared, only the weights differ
Pending-proposal badge	Polled every 30 s

14.4 Test coverage

AREA	TESTS	NOTABLE
Cycle enumeration and optimisation	18	Differential check against an independent brute-force enumerator over 500 random digraphs; 100-run tie-break determinism
Proposal API	17	Concurrent creation yields exactly one success and one conflict
Proposal service	8	Reservation only on last acceptance; cancel releases; expiry frees the donor
Match API	8	Two-way swap end to end; exclusions; audit row written
Graph construction	5	ABO incompatibility and strong DSA both halt
Explanations	2	All four verdicts produced from one pool
Frontend	34	Graph stays desktop-only; workflow badges never use the clinical palette
Total	92	Within 831 across the system, all passing

The database-level uniqueness constraint has its own test that *bypasses the application check entirely* and inserts conflicting rows directly, confirming the guarantee holds independently of the code above it.

REPRODUCING THE WORKED EXAMPLE

The policy comparison in §7 regenerates from a committed script with a fixed random seed, requiring no database. It should be re-run against real pool characteristics before any policy is adopted as programme practice.